TIME SERIES ANALYSIS FOR SALES AND DEMAND FORECASTING

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1. Introduction

Accurate demand forecasting is essential for publishers navigating rising production costs, competitive market pressures, and shifting consumer behaviour. Reliable forecasts support decisions on commissioning, print volumes, stock replenishment, and long-term investment planning. This project applies **time series analysis** to weekly sales data from the industry's leading source of point-of-sale book sales information—to generate actionable insights for publishers.

The organization is exploring a new analytics service aimed at small and medium-sized independent publishers. These businesses often lack in-house forecasting capabilities but require data-driven guidance on expected sales trajectories, seasonal patterns, and the commercial lifespan of titles. Understanding which books exhibit stable demand, recurring seasonality, or long-term resilience can significantly reduce financial risk and improve procurement and reprinting decisions.

Using historical sales data for selected ISBNs, this project applies classical statistical models, machine learning, deep learning, and hybrid approaches to evaluate forecasting performance. The analysis focuses on two titles—*The Alchemist* and *The Very Hungry Caterpillar*—chosen for their data completeness and clear seasonal behaviour. The objective is to identify the most reliable forecasting method for supporting inventory planning, restocking, and long-term investment decisions.

2. Methodology Overview

A multi-model forecasting framework was used to compare classical, machine learning, deep learning, and hybrid approaches. The workflow included data preparation, exploratory analysis, model development, and performance evaluation.

Data Preparation and Initial Investigation

- The dataset consisted of weekly sales observations by ISBN.
- Missing weeks were filled with zeros to maintain continuity, ISBNs were converted to strings, and dates were transformed into datetime format. Titles with sales beyond July 2024 were retained to ensure sufficient recent history.
- Exploratory analysis highlighted differences between early-life and mature-life sales patterns, informing model selection.
- Two titles were chosen for detailed modelling, with analysis restricted to data from 2012 onwards to avoid early lifecycle distortions.

3. Classical Techniques (Auto ARIMA)

Decomposition

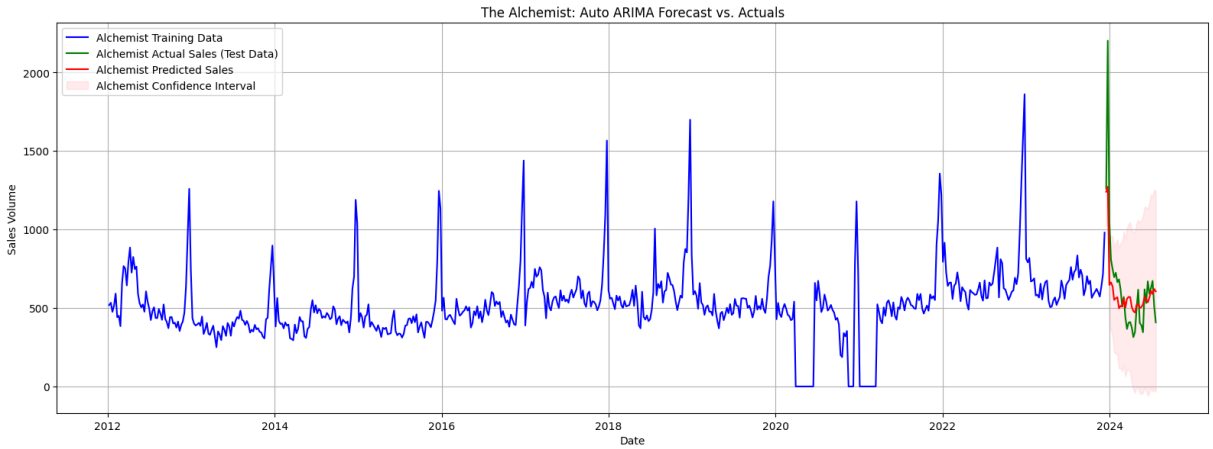
- The decomposition confirms that sales for The Alchemist are driven primarily by stable trend growth and strong annual seasonality, with limited irregular volatility except during external disruptions.
- The clear seasonal structure and trend persistence justify the application of SARIMA modelling as an appropriate classical forecasting framework.

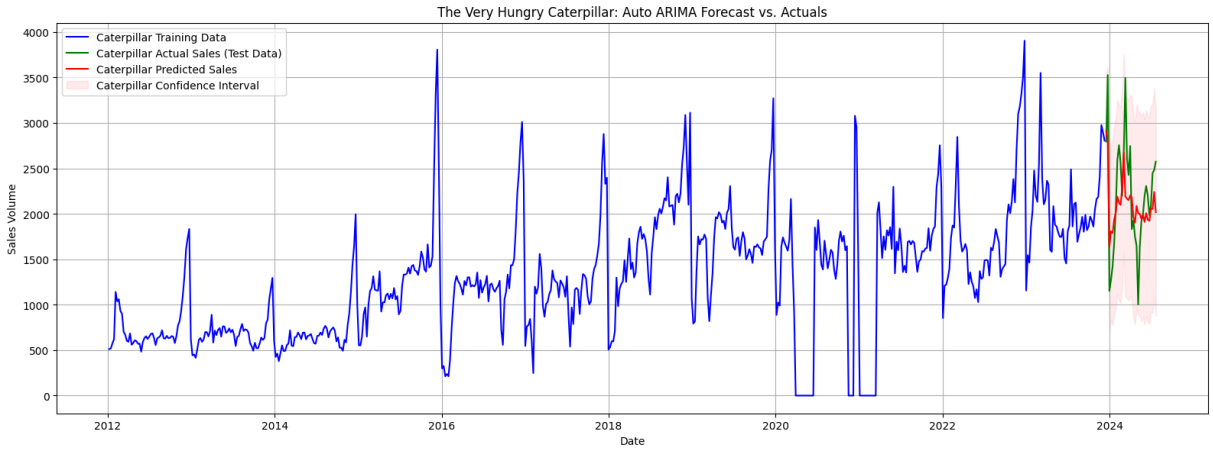
Stationarity

- The ADF test results indicate that both The Alchemist and The Very Hungry Caterpillar are stationary at the 5% significance level, suggesting no strong long-term stochastic trend in either series.
- The Alchemist is strongly stationary in its original form and requires minimal differencing, while The Very Hungry Caterpillar benefits from log transformation and light differencing to improve stability.
- Overall, demand for both titles fluctuates around a stable mean, supporting the use of ARIMA-type models without heavy non-seasonal differencing.

Auto-ARIMA

- Best models: SARIMA(0,1,2)(1,0,1)[52] for The Alchemist and SARIMA(1,1,1)(1,0,1)[52] for The Hungry Caterpillar.
- Both models confirm strong annual seasonality through seasonal AR and MA components.
- The Alchemist exhibits relatively stable demand dynamics, primarily influenced by short-term shock effects, while The Hungry Caterpillar demonstrates stronger autoregressive behaviour and slightly higher structural variability.
- Neither series required seasonal differencing, indicating stable recurring yearly demand patterns.
- Overall, the models suggest predictable seasonality with moderate short-term fluctuations, supporting reliable medium-term forecasting.


The Alchemist: Auto ARIMA Forecast vs. Actuals
Alchemist Training Data
Alchemist Actual Sales (Test Data)
Alchemist Predicted Sales
Alchemist Confidence Interval
Sales Volume
Date


The Very Hungry Caterpillar: Auto ARIMA Forecast vs. Actuals
Caterpillar Training Data
Caterpillar Actual Sales (Test Data)
Caterpillar Predicted Sales
Caterpillar Confidence Interval
Sales Volume
Date

4. Machine Learning & Deep Learning Techniques

XGBoost (Machine Learning)

XGBoost was used to capture nonlinear relationships using engineered features such as lag variables, rolling statistics, and calendar indicators. Hyperparameters were tuned using cross-validation.

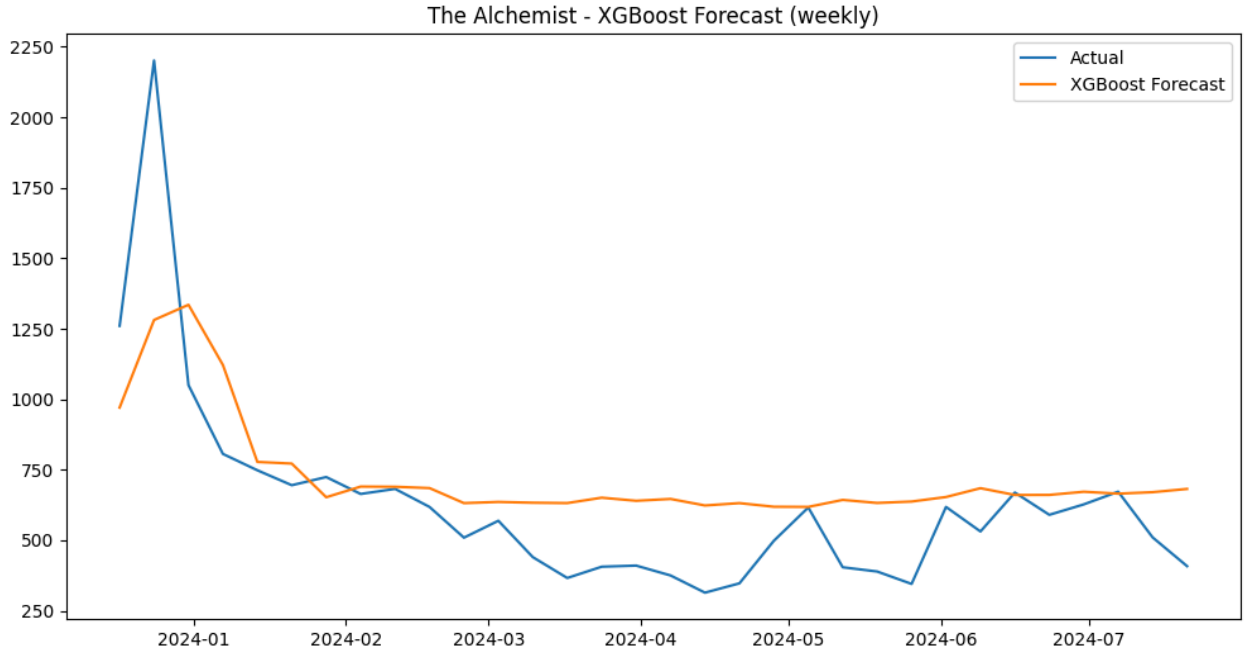
For *The Alchemist*:

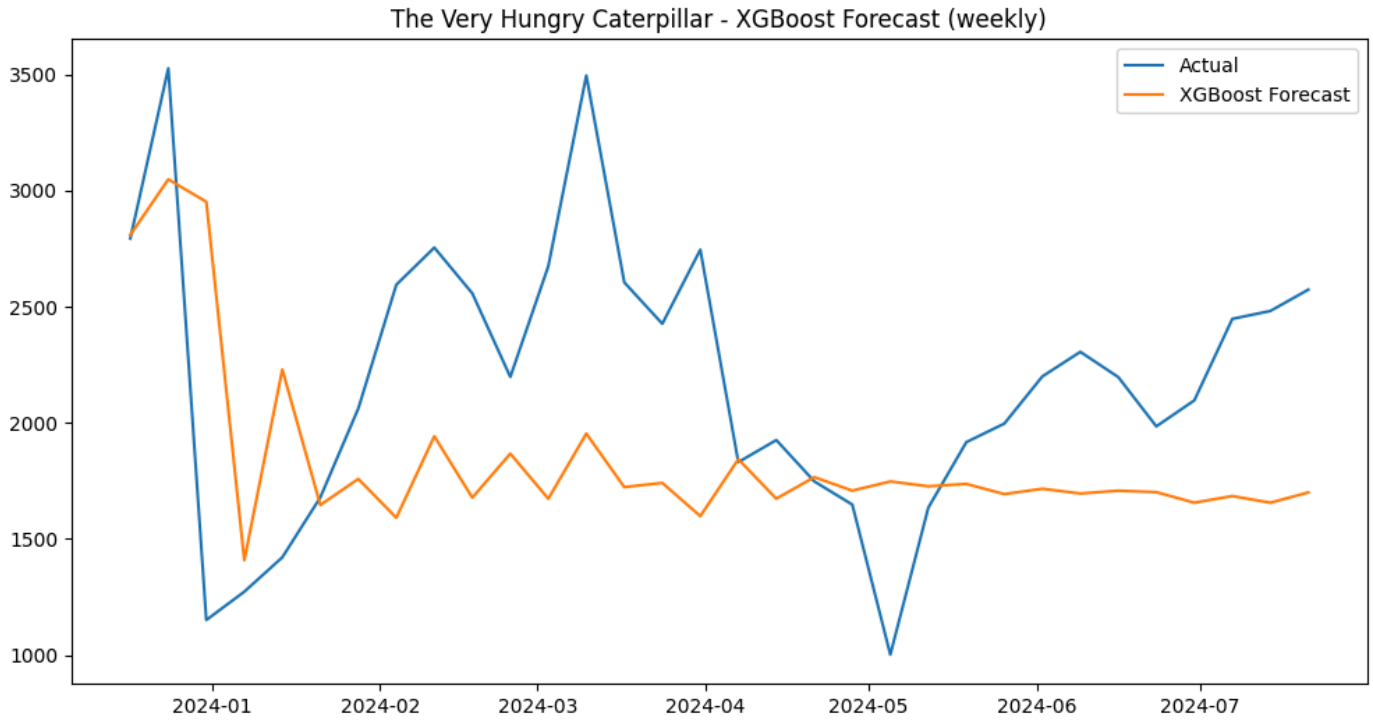
- Forecasts were stable and suitable for baseline inventory planning.
- The model tended to underestimate demand spikes, limiting its usefulness for promotional forecasting.

For *The Very Hungry Caterpillar*:

- XGBoost produced conservative forecasts that reduced overstocking risk.
- However, it struggled with high-demand spikes driven by external factors such as school cycles or promotions.

Overall, XGBoost captured general direction but smoothed extreme fluctuations. It performed reasonably well but showed reduced accuracy on volatile series.





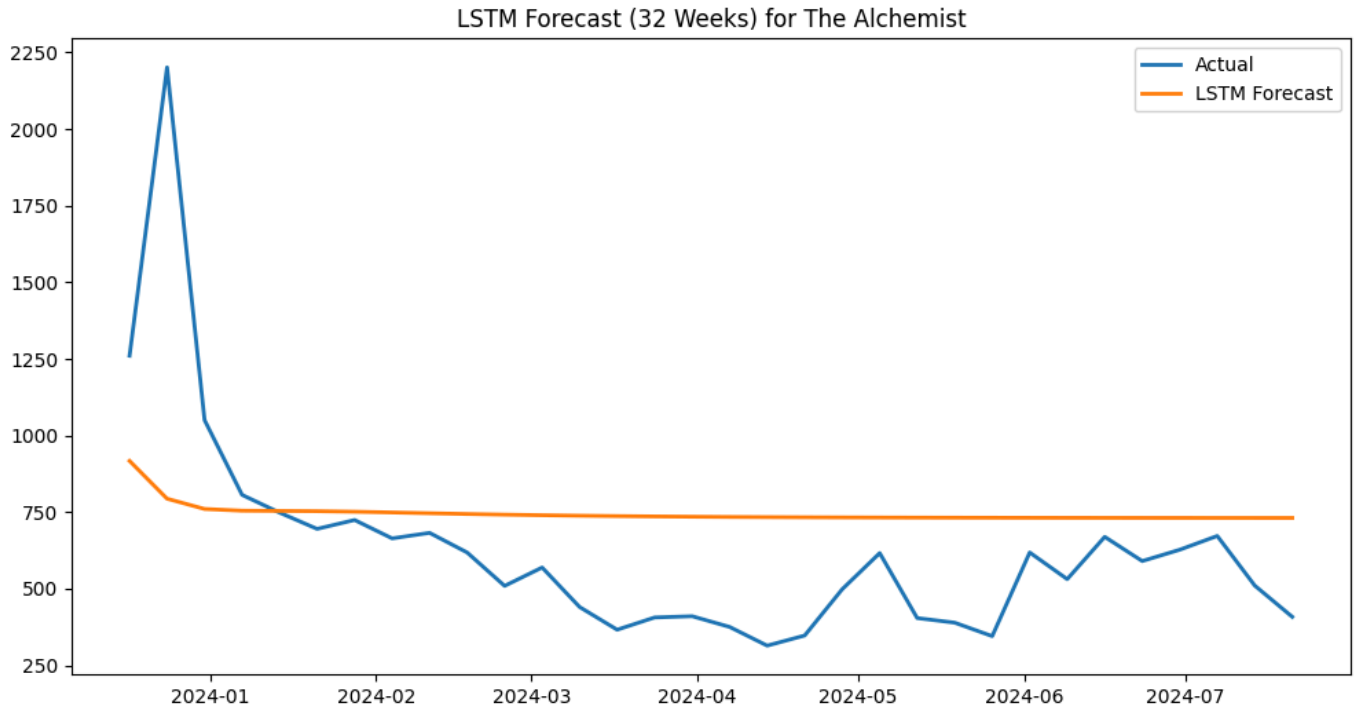
LSTM (Deep Learning)

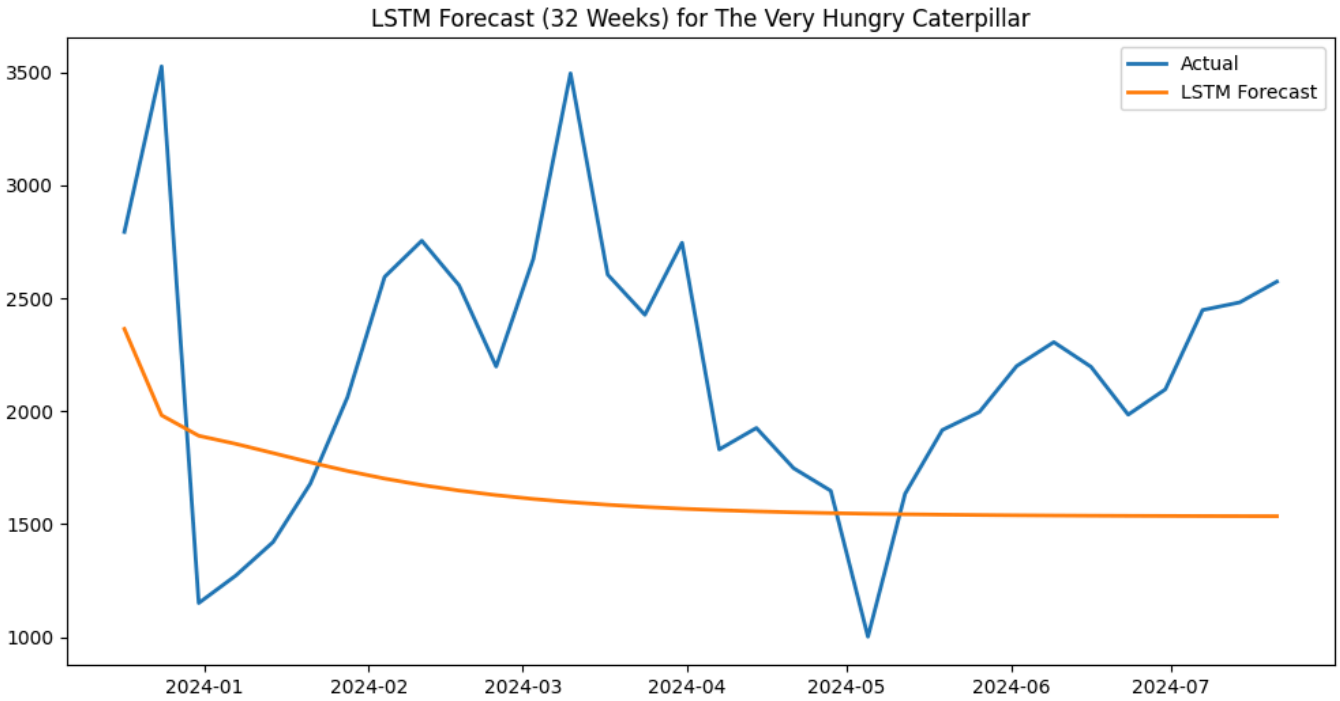
An LSTM neural network was implemented to model sequential dependencies using sliding time windows.

Across both titles:

- LSTM did not outperform XGBoost or SARIMA.
- It introduced additional complexity without improving accuracy.
- High-variance spikes—especially for *The Very Hungry Caterpillar*—were not predictable from historical data alone.
- Deep learning was not justified given dataset size and signal characteristics.

LSTM proved to be the weakest standalone model in this context.


LSTM Forecast (32 Weeks) for The Alchemist
Actual
LSTM Forecast
2250
2000
1750
1500
1250
1000
750
500
250
2024-01
2024-02
2024-03
2024-04
2024-05
2024-06
2024-07


LSTM Forecast (32 Weeks) for The Very Hungry Caterpillar
Actual
LSTM Forecast
3500
3000
2500
2000
1500
1000
2024-01
2024-02
2024-03
2024-04
2024-05
2024-06
2024-07

5. Hybrid Model (SARIMA & LSTM - Sequential and Parallel Combination)

To combine linear structure with nonlinear refinement, two hybrid approaches were tested:

1. **Sequential Hybrid**: SARIMA models the structured components first; LSTM models the residuals.
2. **Parallel Hybrid**: SARIMA and LSTM are trained independently and their forecasts averaged.

Hybrid Model Performance

The **sequential SARIMA-LSTM hybrid** delivered the strongest performance across both titles, significantly reducing forecast error compared to standalone models.

- For *The Very Hungry Caterpillar*, the improvement was substantial due to its volatility.
- The hybrid approach balanced stability (SARIMA) with responsiveness (LSTM).
- Parallel hybrid models improved accuracy but were less effective than sequential models.

Why Hybrid Works Best

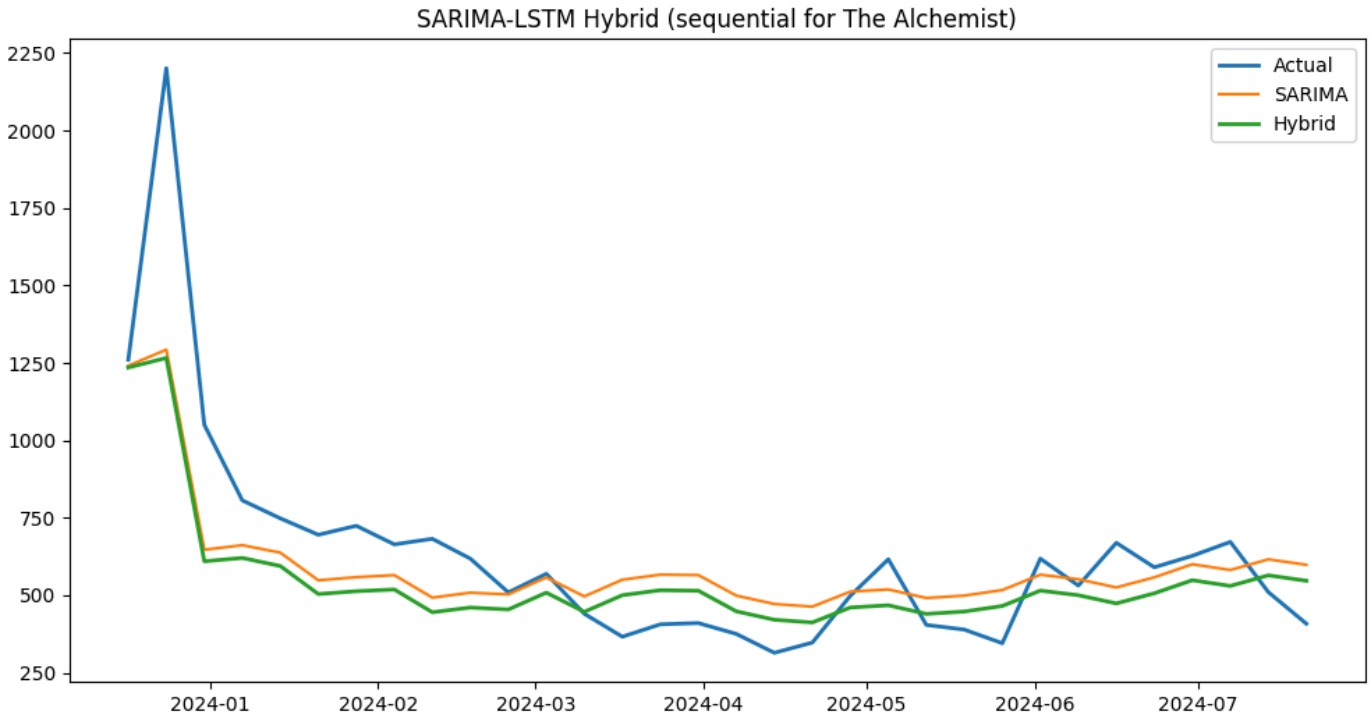
Book sales contain:

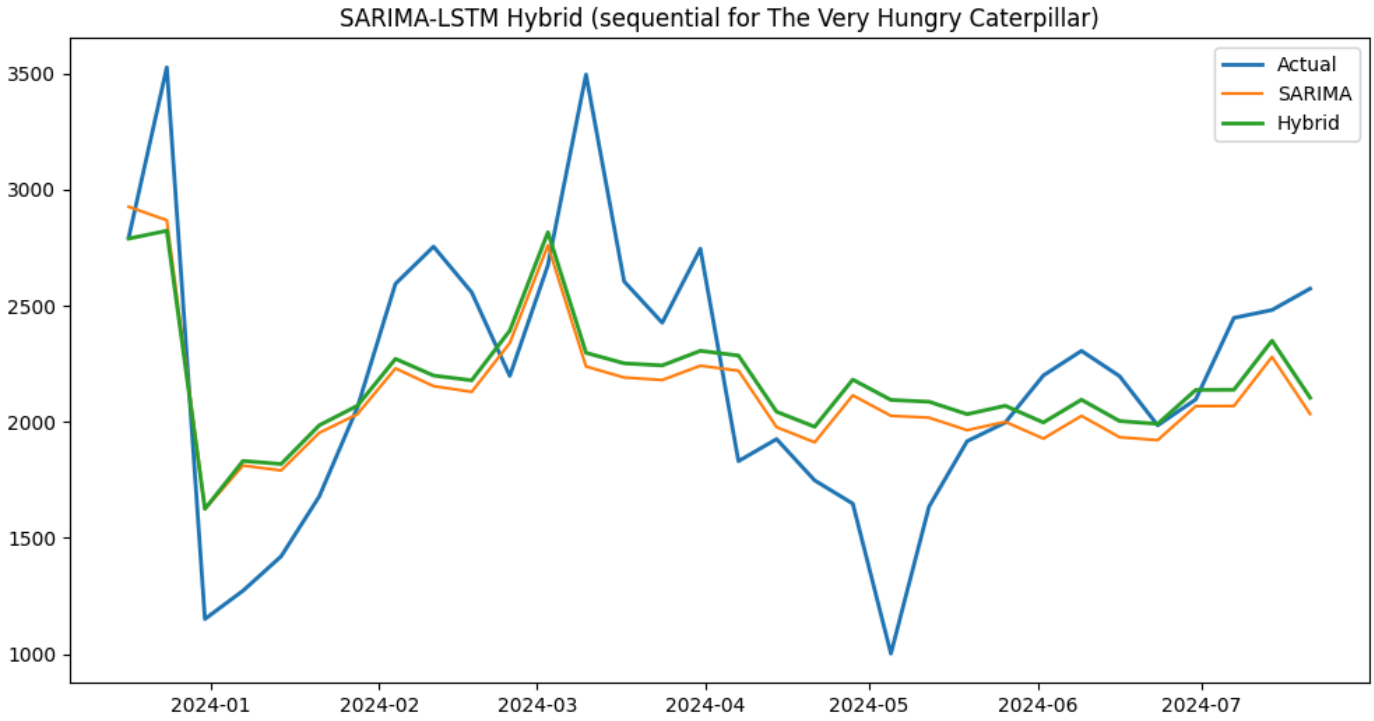
- Structural seasonal components
- Linear long-term trends
- Nonlinear short-term fluctuations

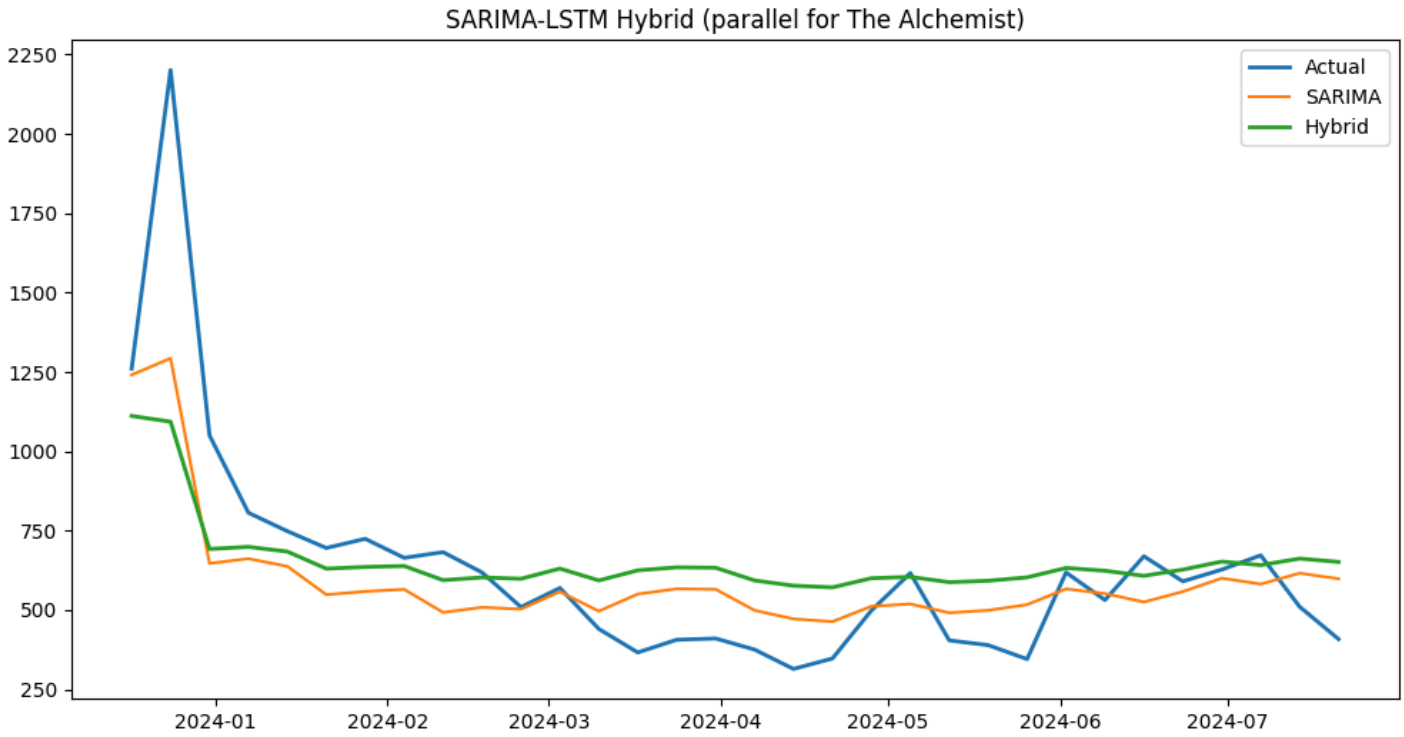
No single model captures all components effectively. The sequential hybrid succeeds because:

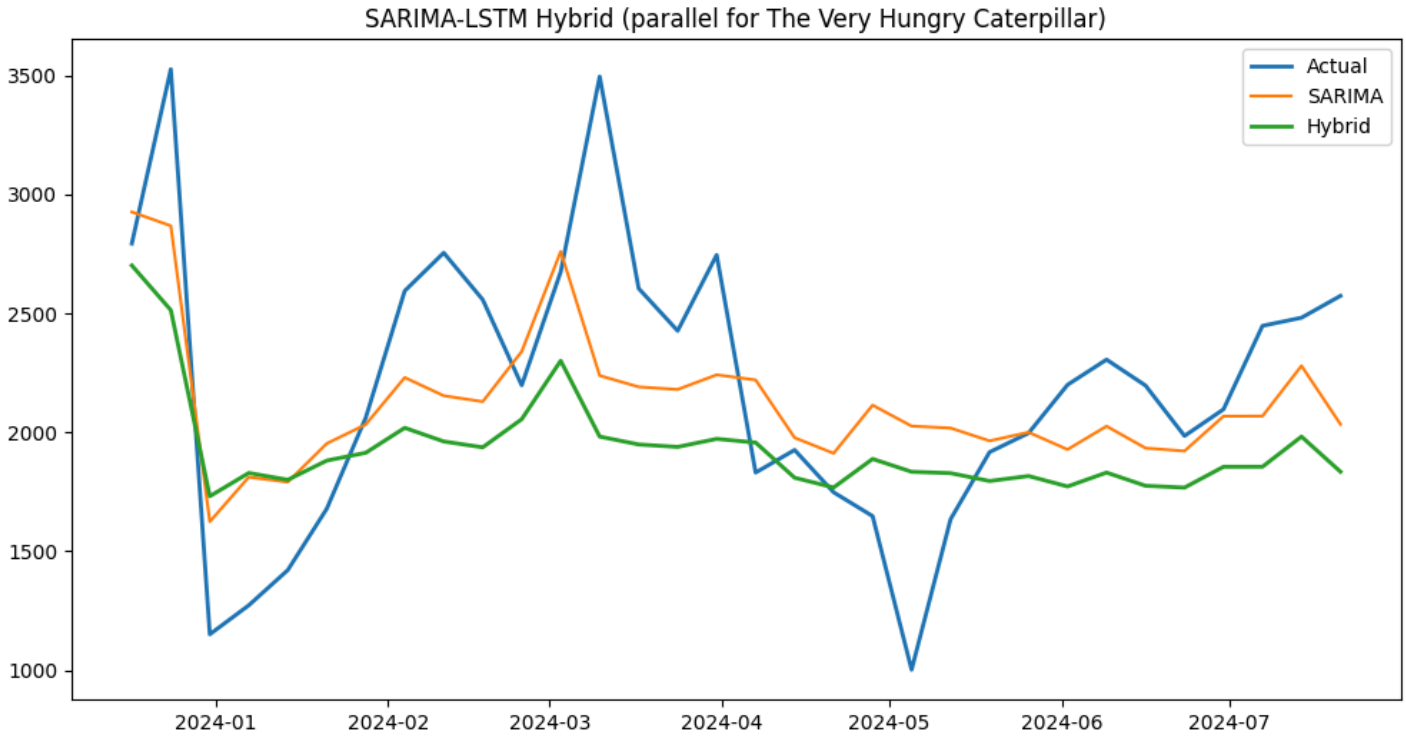
1. SARIMA captures predictable structure.
2. LSTM models remaining irregular patterns.

This layered approach reduces both systematic and volatility-related error, making it the most commercially reliable forecasting method.


SARIMA-LSTM Hybrid (sequential for The Alchemist)
Actual
SARIMA
Hybrid
2250
2000
1750
1500
1250
1000
750
500
250
2024-01
2024-02
2024-03
2024-04
2024-05
2024-06
2024-07


SARIMA-LSTM Hybrid (sequential for The Very Hungry Caterpillar)
Actual
SARIMA
Hybrid
3500
3000
2500
2000
1500
1000
2024-01
2024-02
2024-03
2024-04
2024-05
2024-06
2024-07


SARIMA-LSTM Hybrid (parallel for The Alchemist)
Actual
SARIMA
Hybrid
2250
2000
1750
1500
1250
1000
750
500
250
2024-01
2024-02
2024-03
2024-04
2024-05
2024-06
2024-07


SARIMA-LSTM Hybrid (parallel for The Very Hungry Caterpillar)
Actual
SARIMA
Hybrid
3500
3000
2500
2000
1500
1000
2024-01
2024-02
2024-03
2024-04
2024-05
2024-06
2024-07

6. Monthly Prediction

To evaluate the impact of temporal aggregation, monthly forecasts were generated using **SARIMA** and **XGBoost**.

Key Findings

Monthly Data

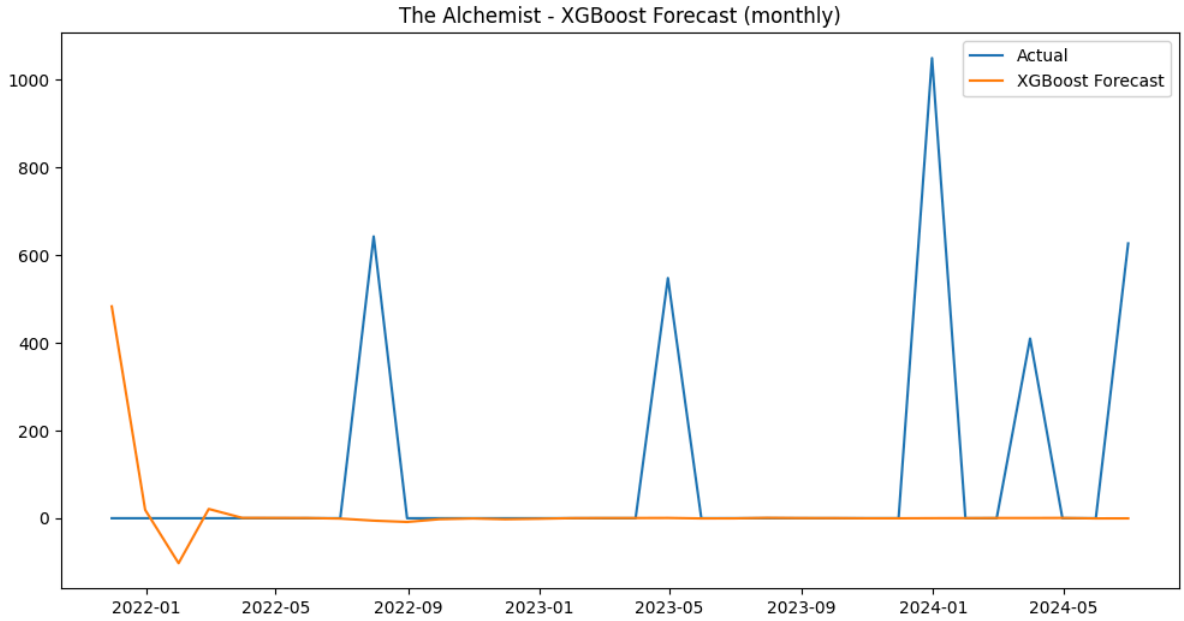
- SARIMA and XGBoost produced similar MAE values.
- MAPE was unreliable for SARIMA due to near-zero sales months.
- XGBoost produced a more stable MAPE but still struggled with sparse months.

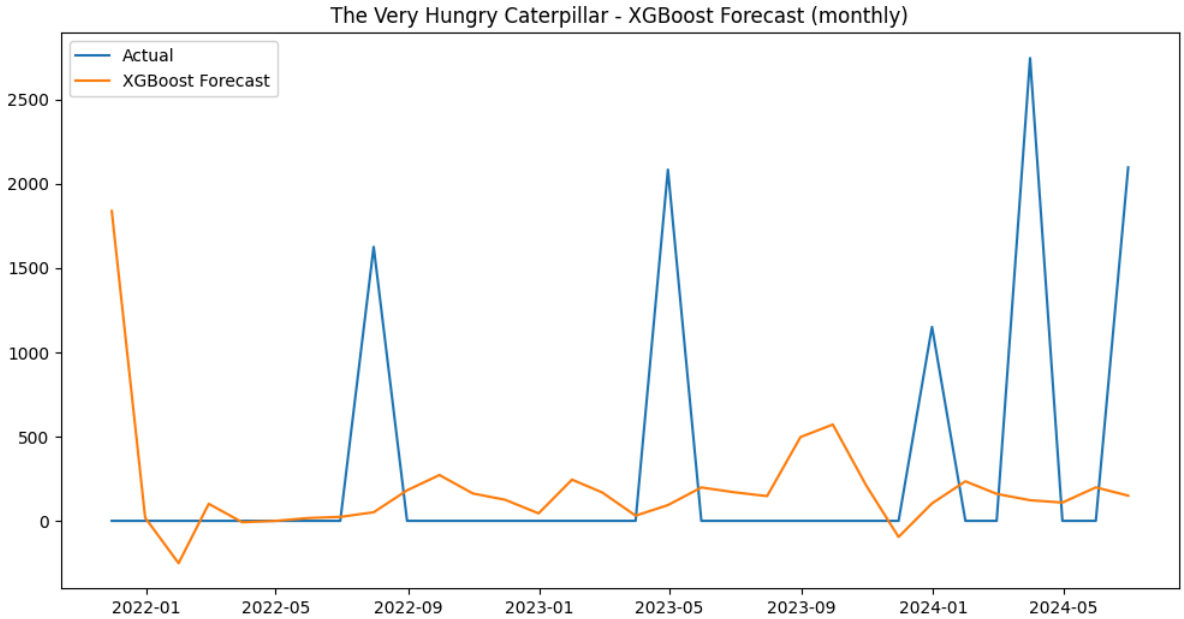
Weekly Data

- SARIMA weekly forecasts were highly accurate (MAPE ~0.2%).
- XGBoost weekly forecasts were moderately accurate (MAPE ~28%).
- SARIMA weekly outperformed XGBoost for *The Very Hungry Caterpillar* and performed similarly for *The Alchemist*.

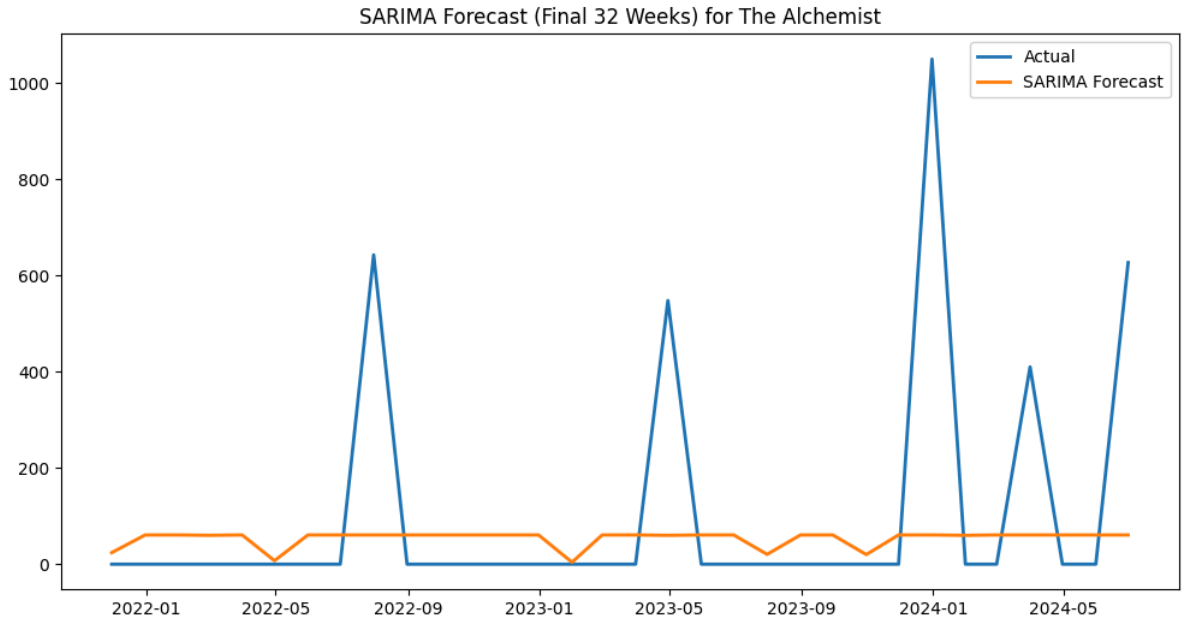
Summary

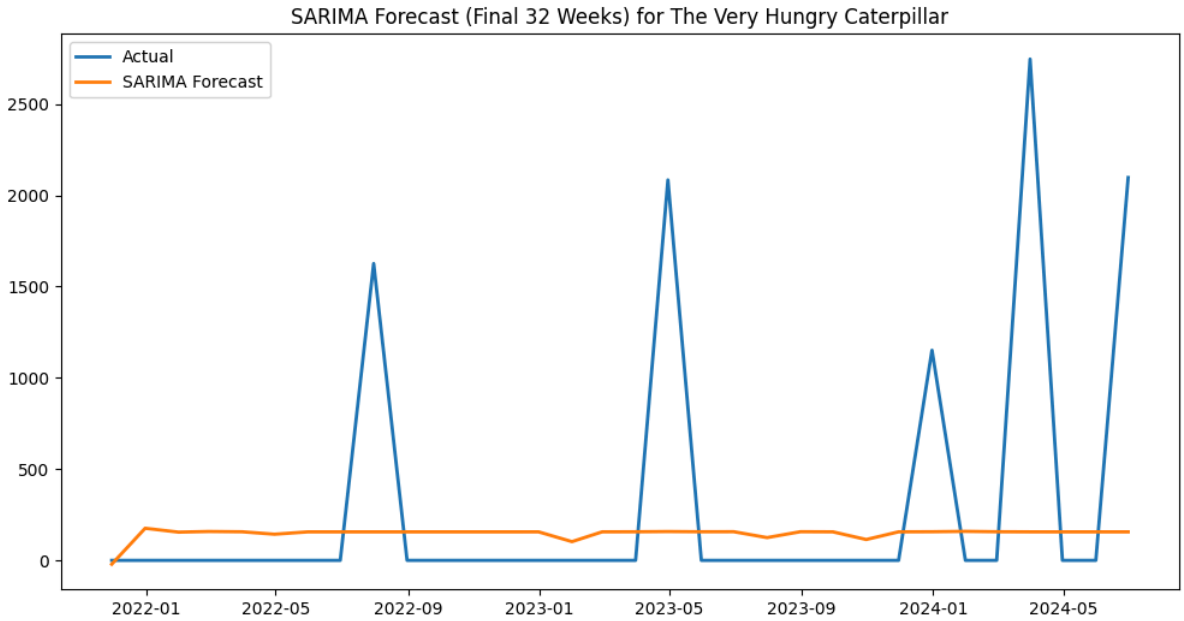
- **Weekly SARIMA is the most accurate** method for proportional trend prediction.
- Monthly forecasts are more suitable for budgeting and strategic planning but less reliable for precise accuracy.
- XGBoost remains useful when external regressors or nonlinear patterns are relevant.


The Alchemist - XGBoost Forecast (monthly)
Actual
XGBoost Forecast
1000
800
600
400
200
0
2022-01
2022-05
2022-09
2023-01
2023-05
2023-09
2024-01
2024-05

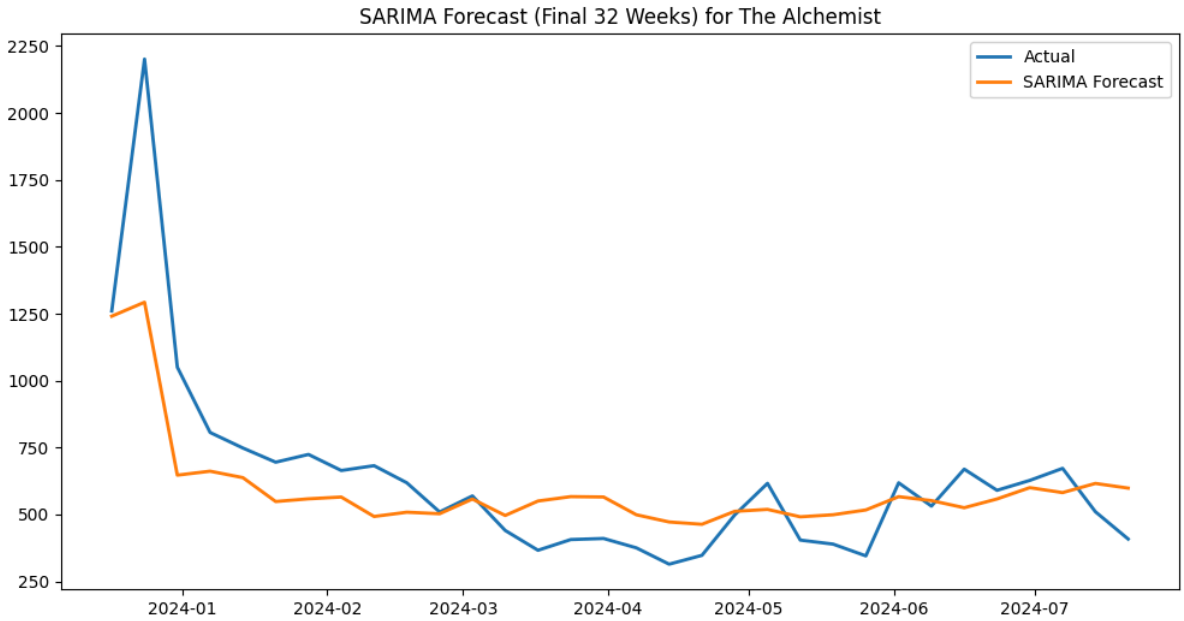

The Very Hungry Caterpillar - XGBoost Forecast (monthly)
Actual
XGBoost Forecast
2500
2000
1500
1000
500
0
2022-01
2022-05
2022-09
2023-01
2023-05
2023-09
2024-01
2024-05

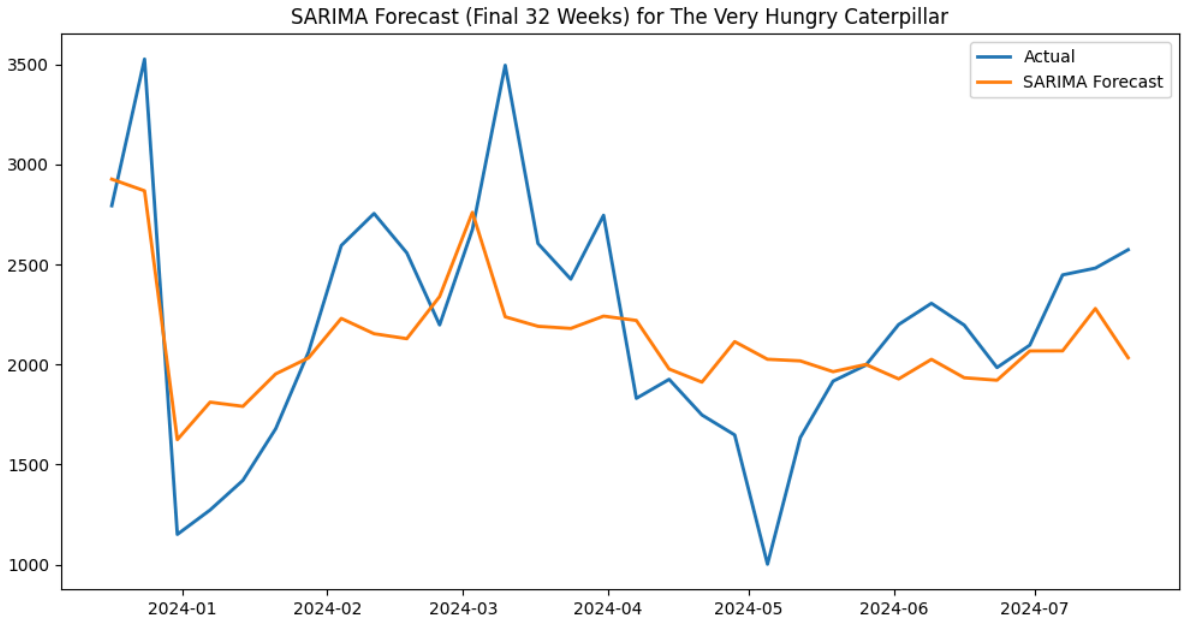
SARIMA - Monthly Forecast Plots





SARIMA - Weekly Forecast Plots





7. Key Business Insights

1. Hybrid Forecasting Is the Most Reliable Approach

   The sequential SARIMA-LSTM model consistently delivered the lowest forecast error across both titles. This confirms that combining structured seasonal modelling with nonlinear refinement provides superior predictive accuracy and commercial value.

2. Demand Characteristics Influence Model Suitability

   - *The Alchemist* shows stable seasonal behaviour and is easier to forecast.
   - *The Very Hungry Caterpillar* is more volatile but still benefits significantly from hybrid modelling.
   - No single standalone model captured both structure and variability effectively.

3. Deep Learning Alone Is Not Sufficient

   LSTM underperformed relative to XGBoost and SARIMA. Dataset size and signal complexity did not justify deep learning as a standalone solution.

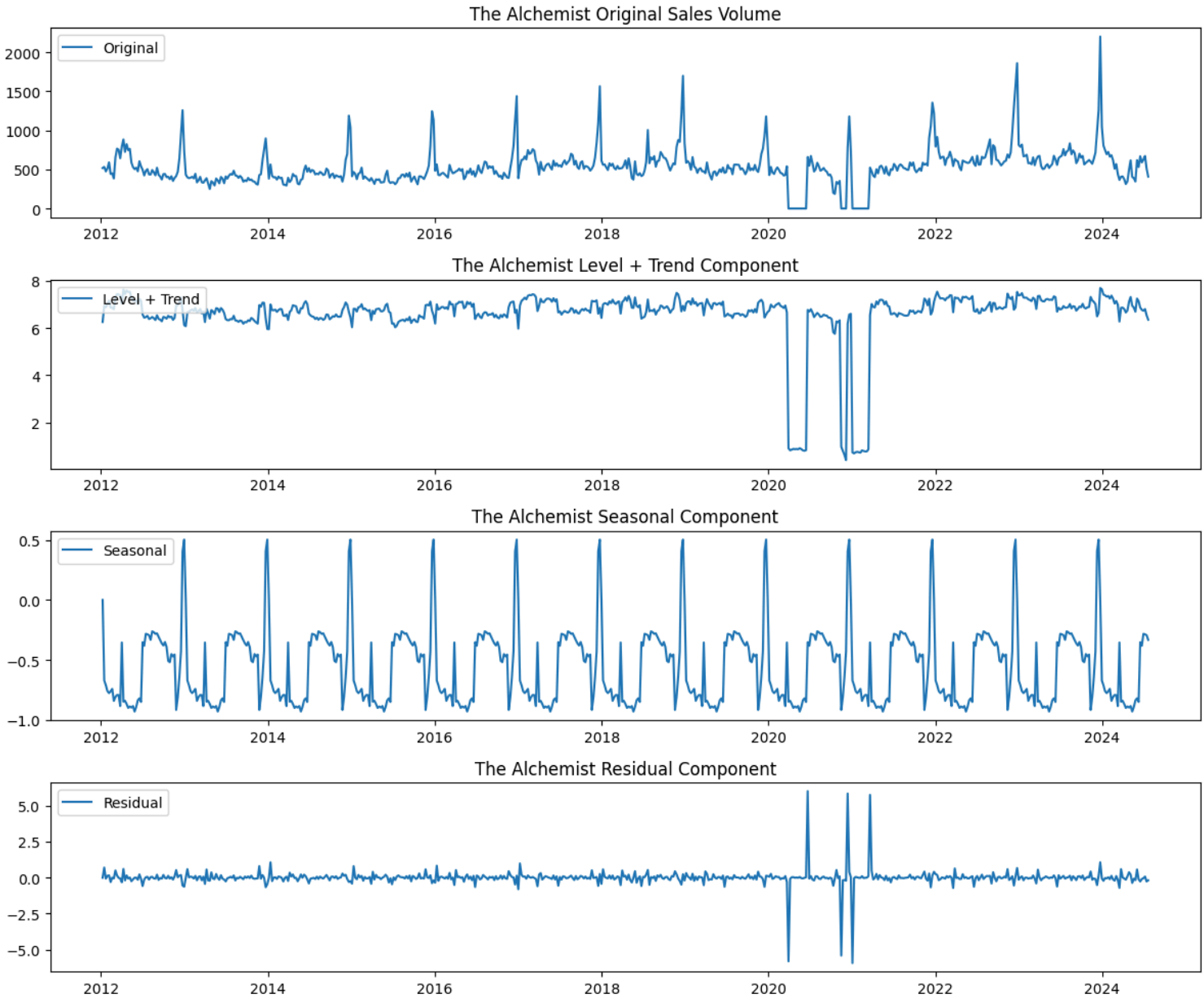
4. Aggregation Supports Strategic Planning

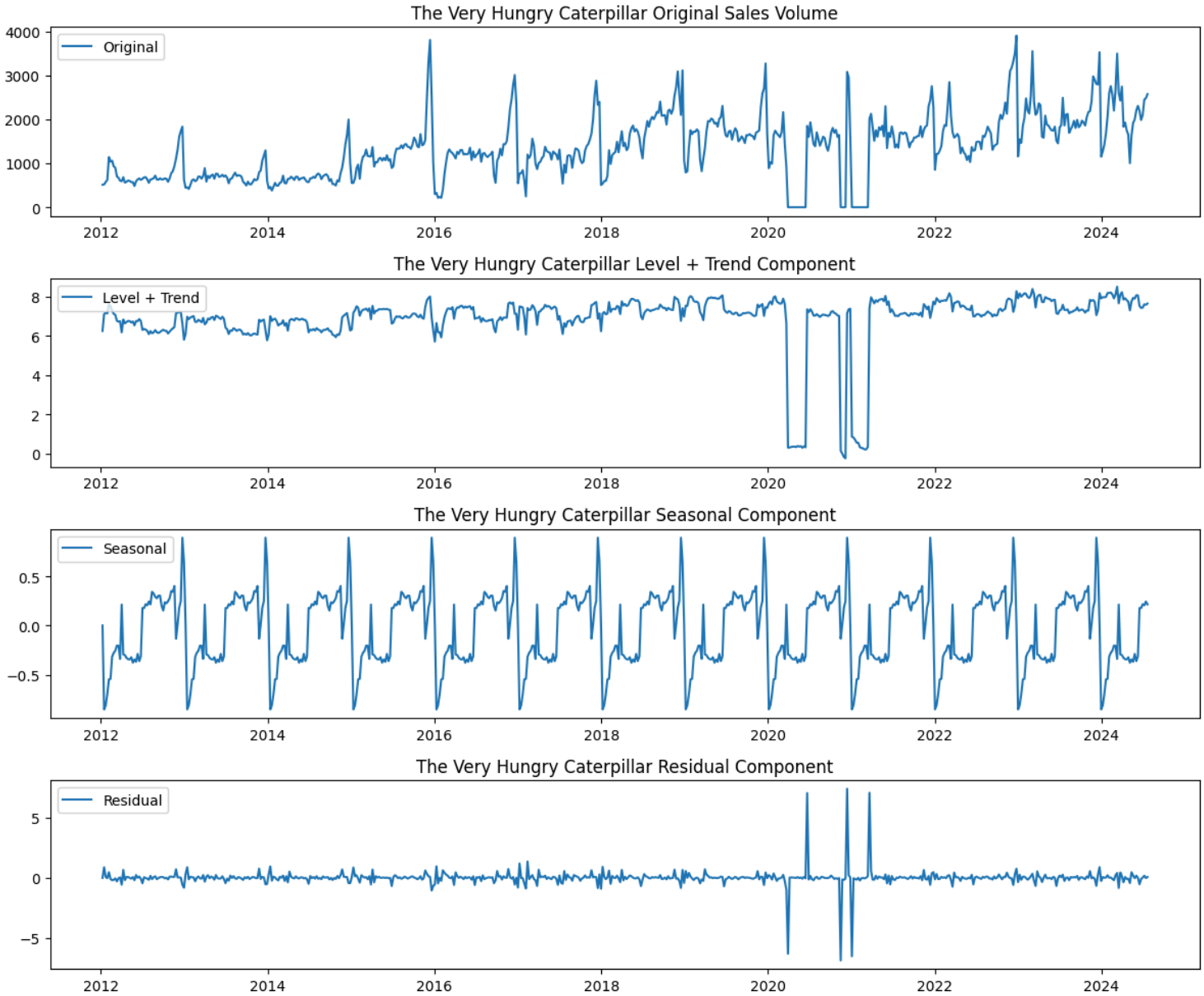
   - Monthly forecasts reduce volatility and support medium-term budgeting.
   - Weekly hybrid forecasts are ideal for operational stock management.

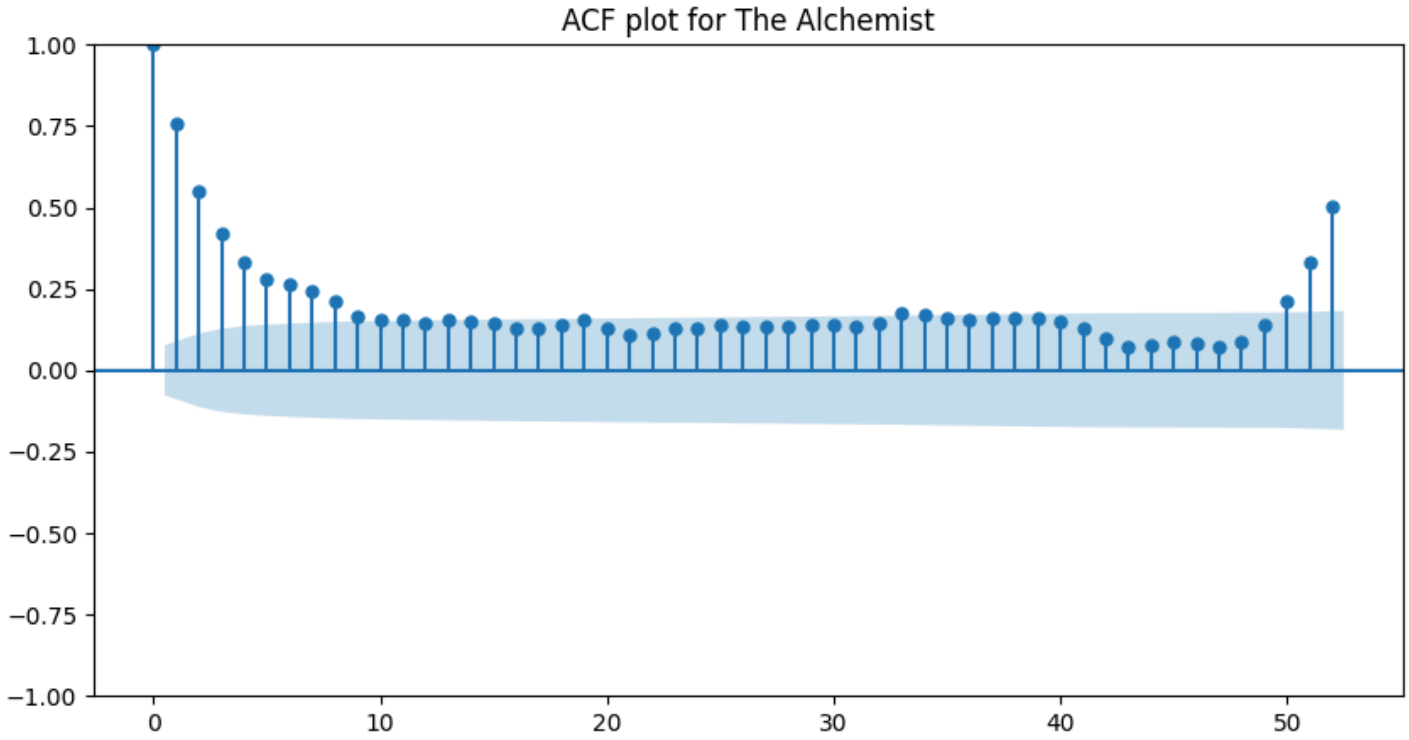
Limitations

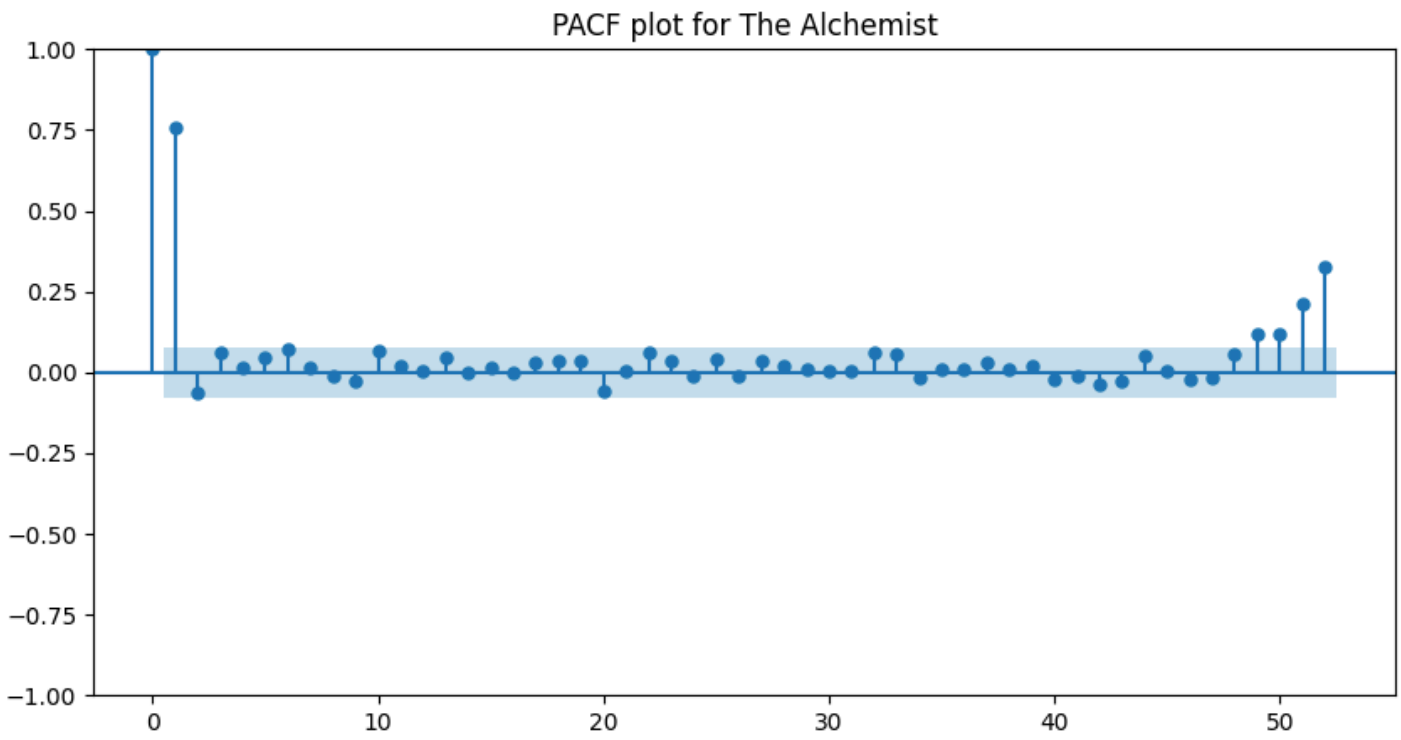
- The analysis relies solely on historical sales and excludes external drivers such as promotions, pricing, and macroeconomic factors.
- Missing values were treated as zero sales, which may not fully reflect reporting gaps.
- Findings are based on two titles and require broader validation for portfolio-wide deployment.

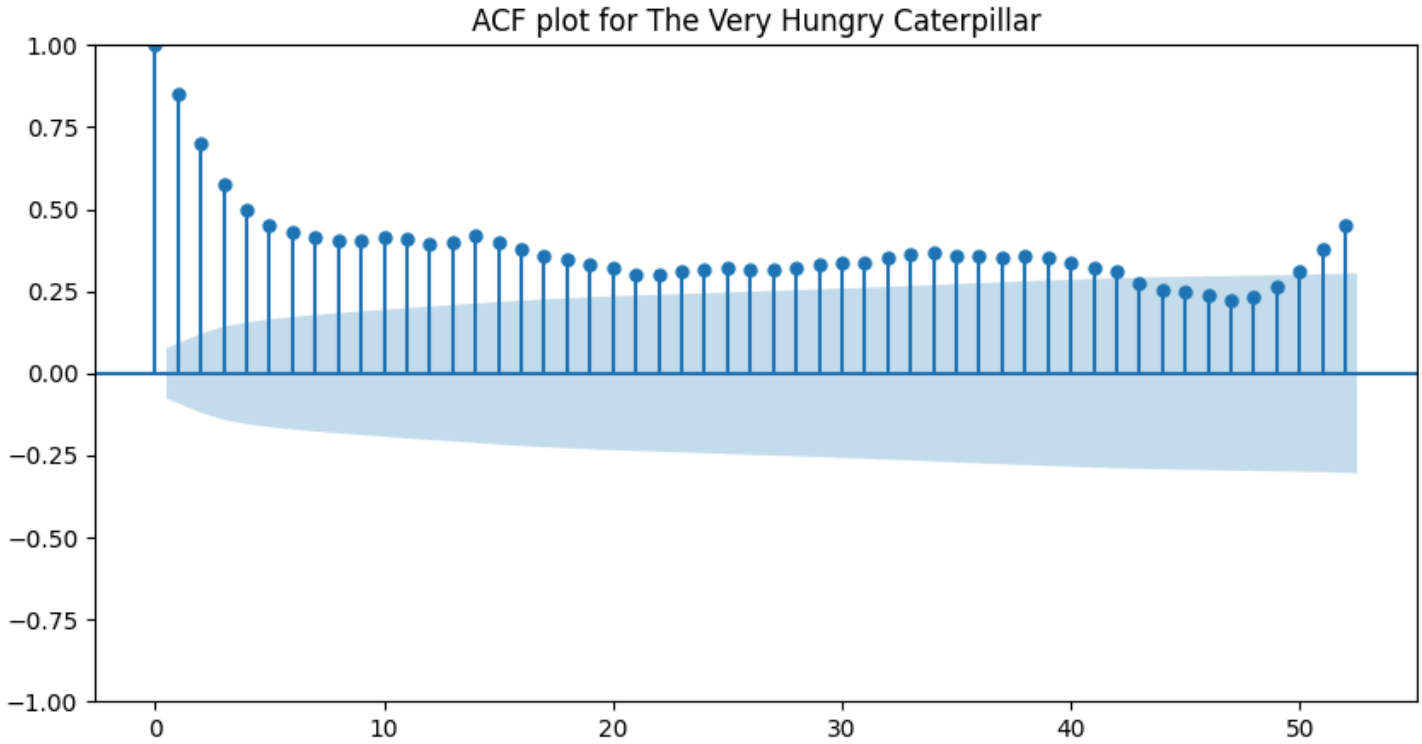
7. Appendix

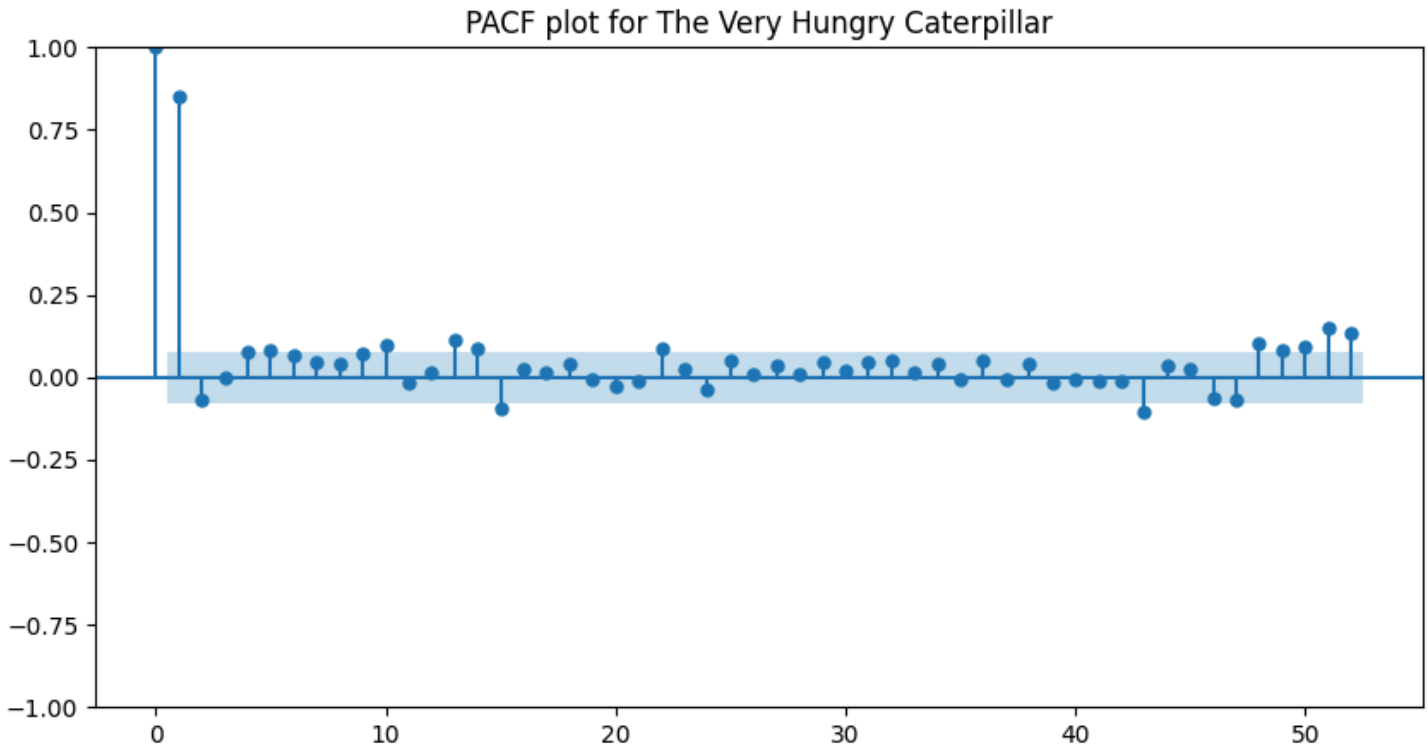

The Alchemist Original Sales Volume
Original
2000
1500
1000
500
0
2012
2014
2016
2018
2020
2022
2024
The Alchemist Level + Trend Component
Level + Trend
8
6
4
2
2012
2014
2016
2018
2020
2022
2024
The Alchemist Seasonal Component
Seasonal
0.5
0.0
−0.5
−1.0
2012
2014
2016
2018
2020
2022
2024
The Alchemist Residual Component
Residual
5.0
2.5
0.0
−2.5
−5.0
2012
2014
2016
2018
2020
2022
2024

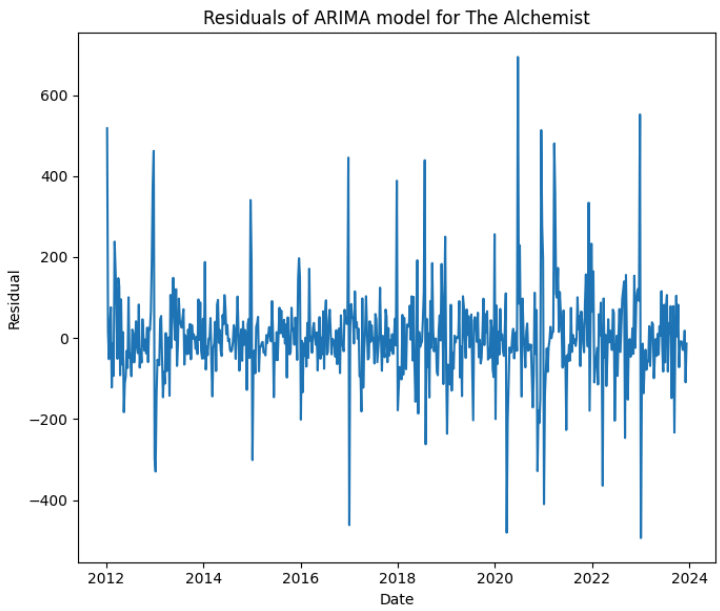

The Very Hungry Caterpillar Original Sales Volume
Original
4000
3000
2000
1000
0
2012
2014
2016
2018
2020
2022
2024
The Very Hungry Caterpillar Level + Trend Component
Level + Trend
8
6
4
2
0
2012
2014
2016
2018
2020
2022
2024
The Very Hungry Caterpillar Seasonal Component
Seasonal
0.5
0.0
−0.5
2012
2014
2016
2018
2020
2022
2024
The Very Hungry Caterpillar Residual Component
Residual
5
0
−5
2012
2014
2016
2018
2020
2022
2024

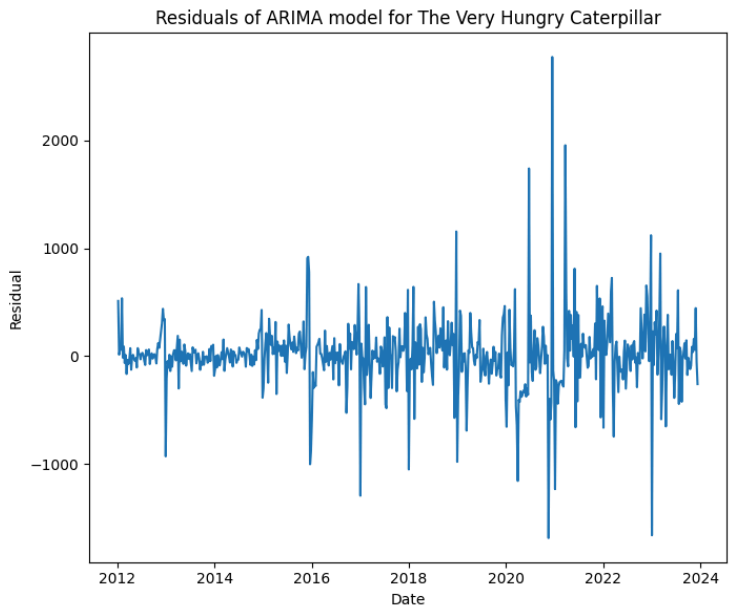

ACF plot for The Alchemist
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
10
20
30
40
50


PACF plot for The Alchemist
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
10
20
30
40
50


ACF plot for The Very Hungry Caterpillar
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
10
20
30
40
50


PACF plot for The Very Hungry Caterpillar
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
10
20
30
40
50


Residuals of ARIMA model for The Alchemist
Residual
Date
600
400
200
0
−200
−400
2012
2014
2016
2018
2020
2022
2024


Residuals of ARIMA model for The Very Hungry Caterpillar
Residual
Date
2000
1000
0
−1000
2012
2014
2016
2018
2020
2022
2024